

Low-Cost Smart Buoy for Near-Shore Wave and Weather Monitoring

Introduction & Motivation

Problem

Near-shore wave data is scarce. Commercial buoys cost **\$10,000+**, limiting accessibility for coastal engineers and surf communities

Gap

Limited low-cost, deployable solutions exist for near-shore wave, wind & weather monitoring in surf zones

Objective

Design, manufacture & deploy a functional smart buoy for **<\$1,000 CAD**

Deployment site

West coast of Vancouver Island (Port Renfrew, BC)

Data collected: Wave height, period & direction · Wind speed & direction, Air & water temperature. All transmitted in near real-time

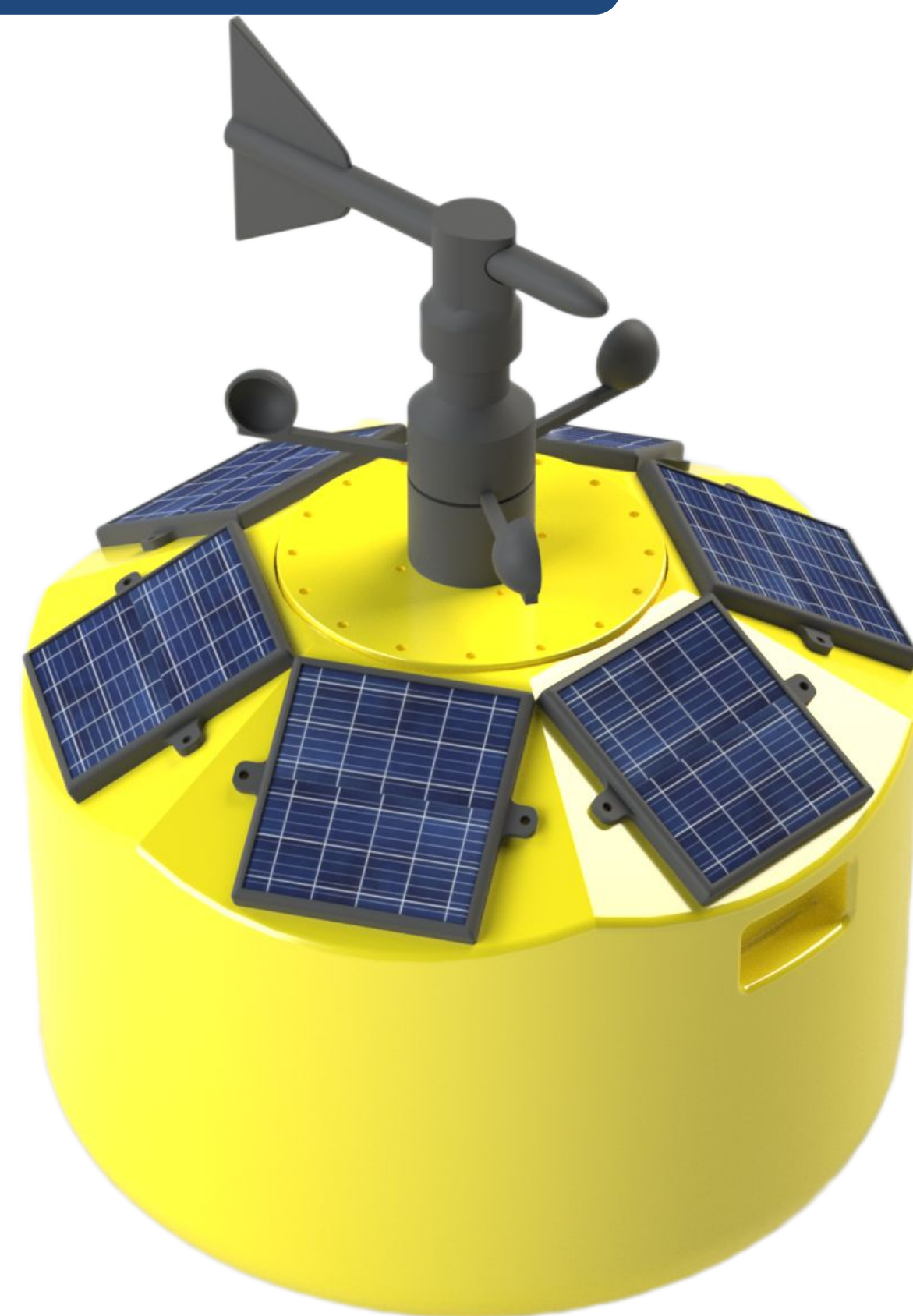
Mechanical Design

Hull & Structure

- ❖ Cylindrical PLA body, printed in segments
- ❖ 2-part epoxy resin coating for IPX-8 waterproofing
- ❖ Access hatch with gasket seal (top face)
- ❖ Mooring eyelet at hull base

Internal Layout

- ❖ **Bottom:** 4× sand compartments
- ❖ **Mid:** LiFePO4 battery enclosure
- ❖ **Upper:** Electronics bay (MCU, sensors)

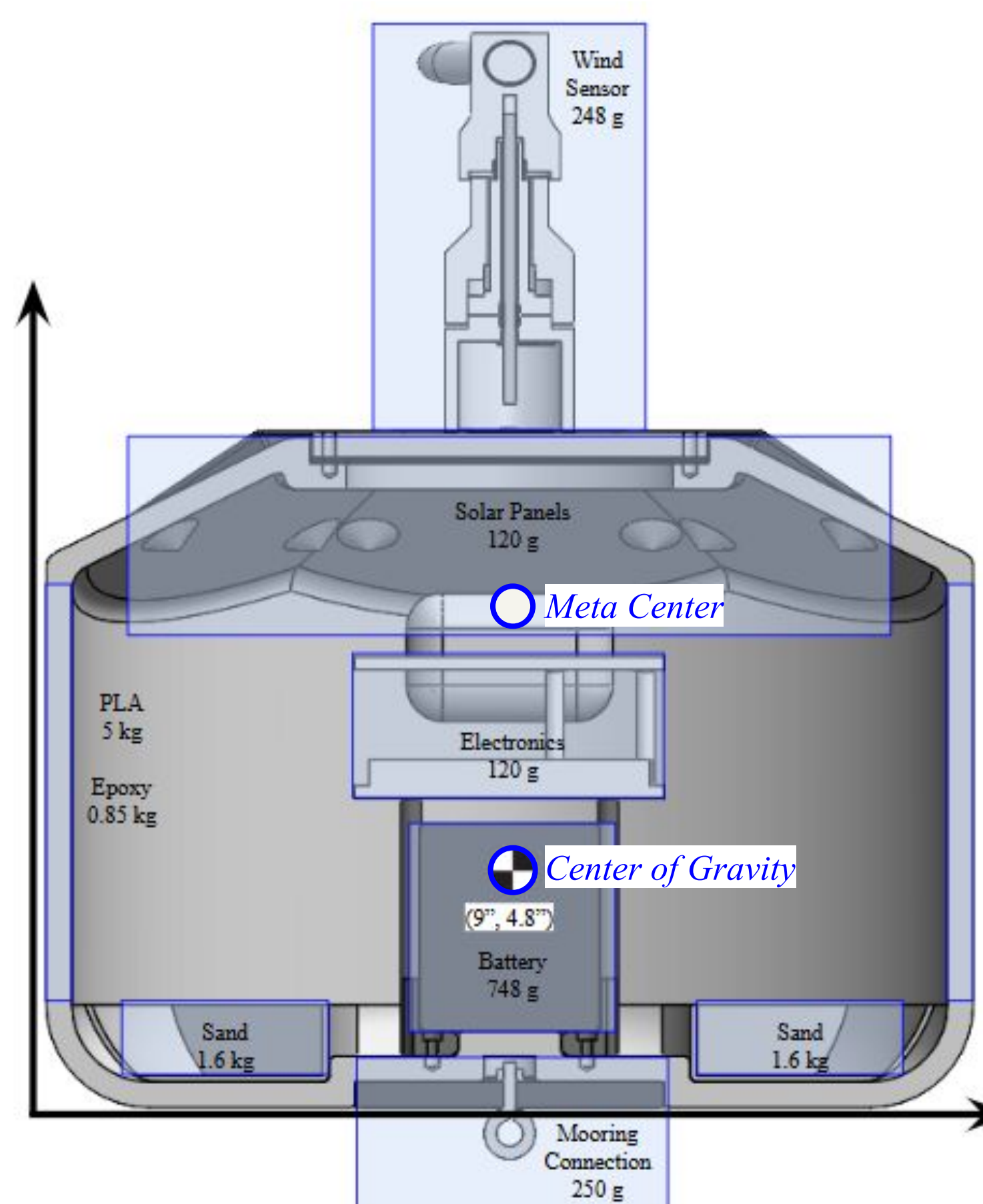


Buoyancy & Stability

10.5 kg
Total system mass

1.23 "
Center of Buoyancy

2.45 "
Draft depth



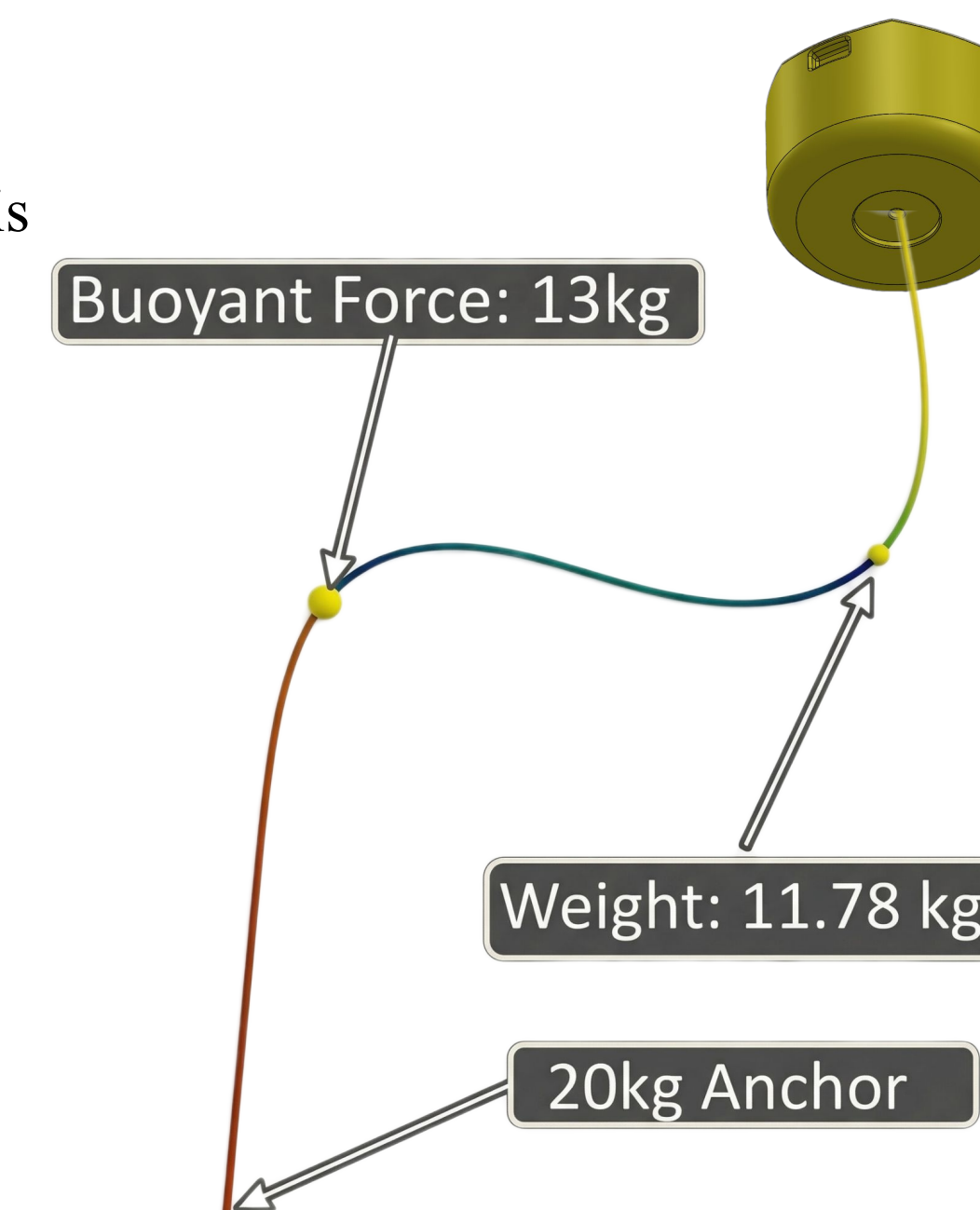
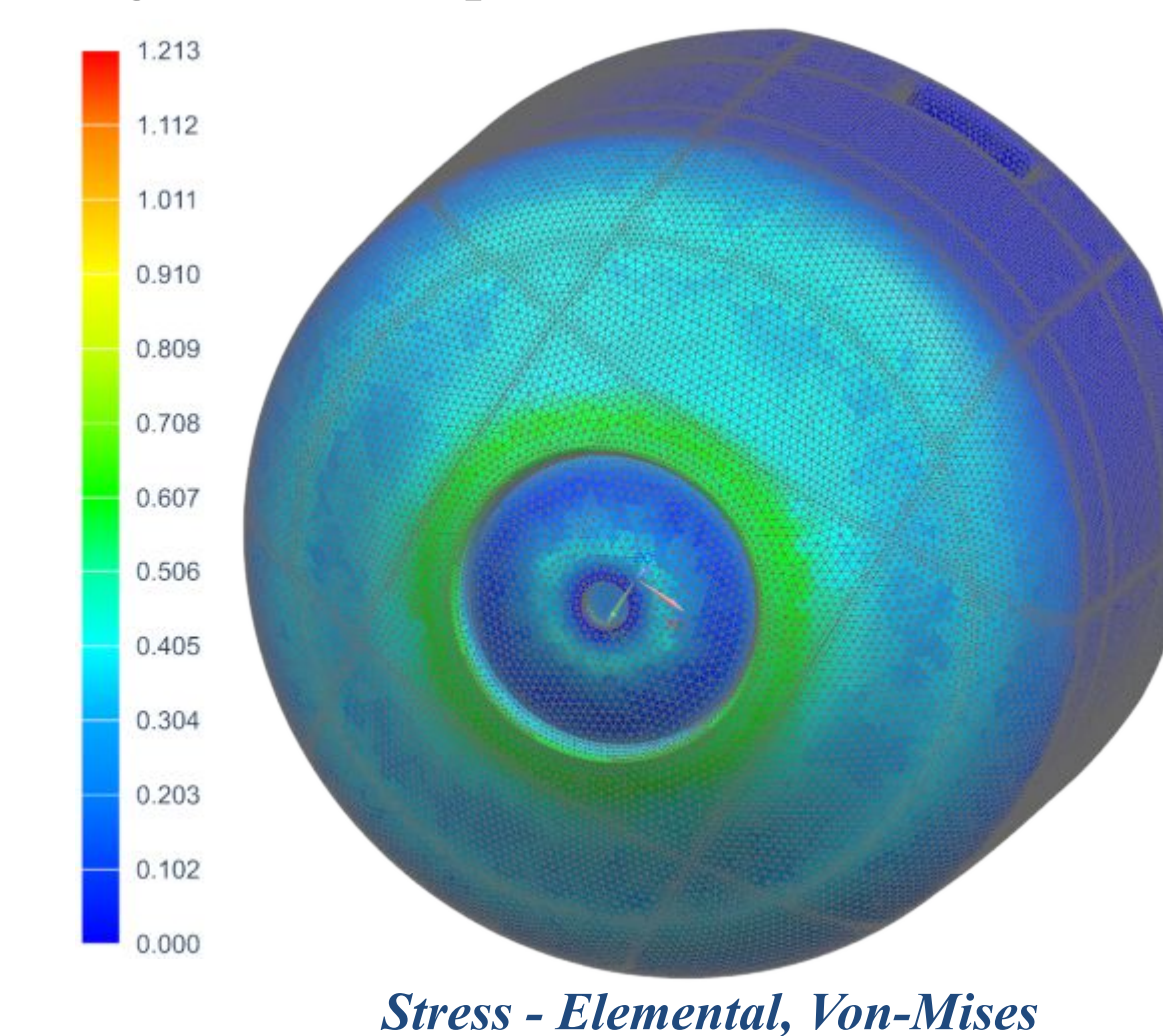
Hydrodynamic Modelling

Issue found: Straight vertical mooring dragged buoy underwater under wave loading, creating higher tension on the bottom of the buoy

Solution: Lazy-S mooring - eliminates high tension forces; validated in simulation

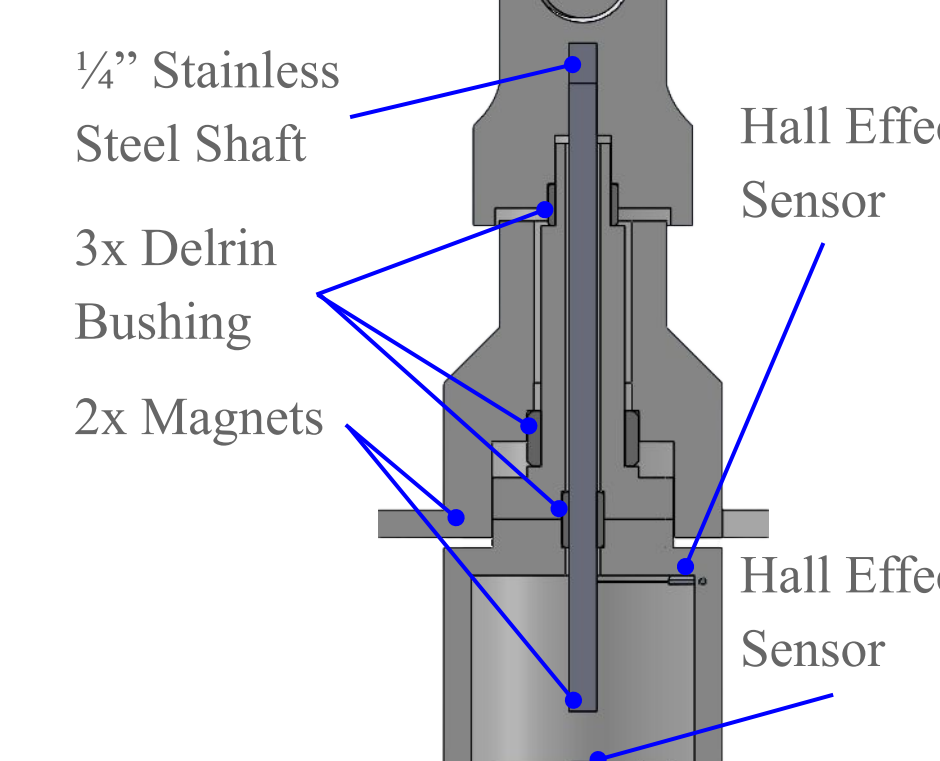
Test conditions:

2.5 m @ 10 s + 0.5 kt tidal current (Port Renfrew realistic)
Worst-case: 5 m waves - buoy remains afloat and attached
Mooring tension output feeds structural FEA loading analysis

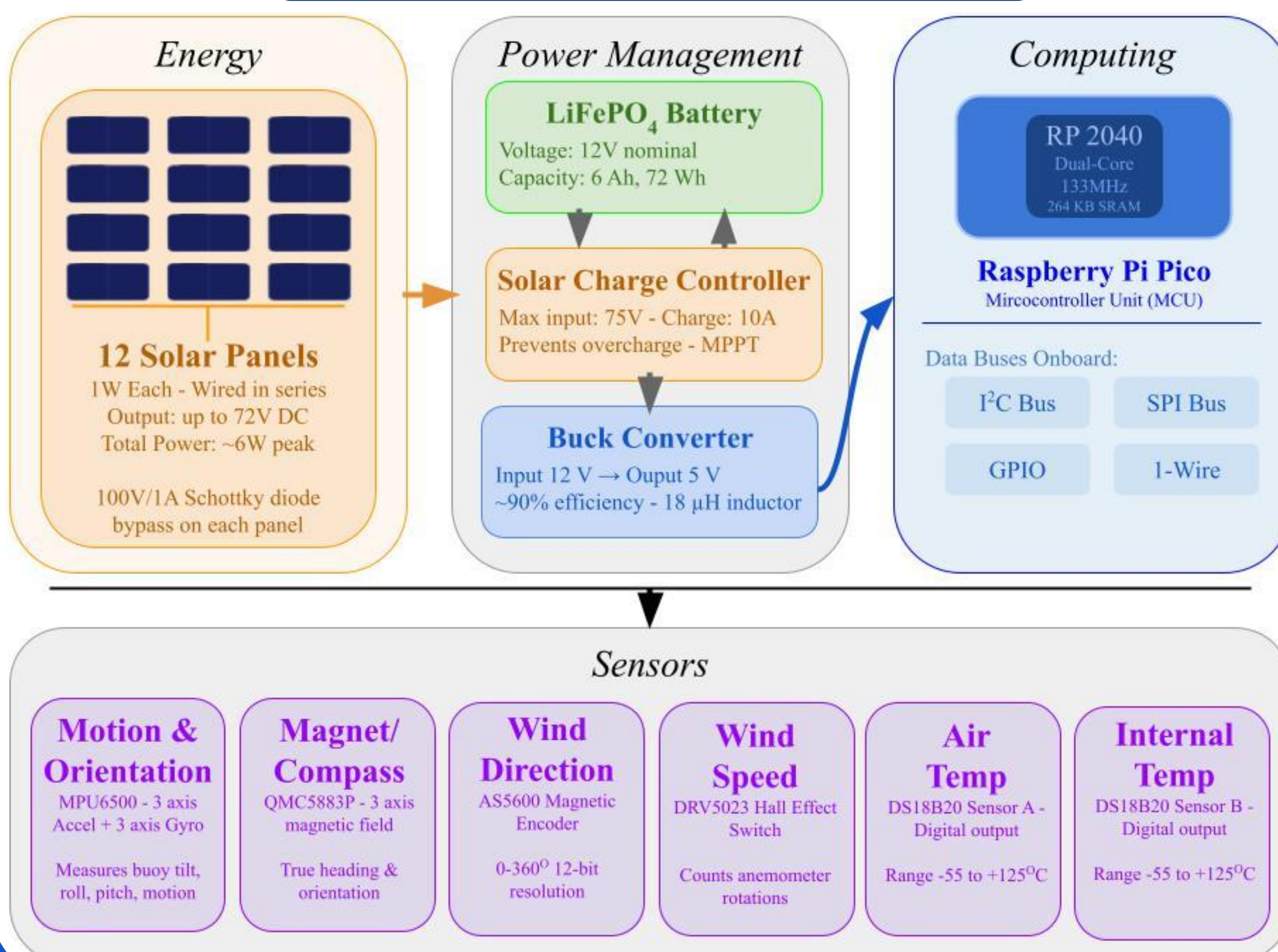


Wind Sensor Design

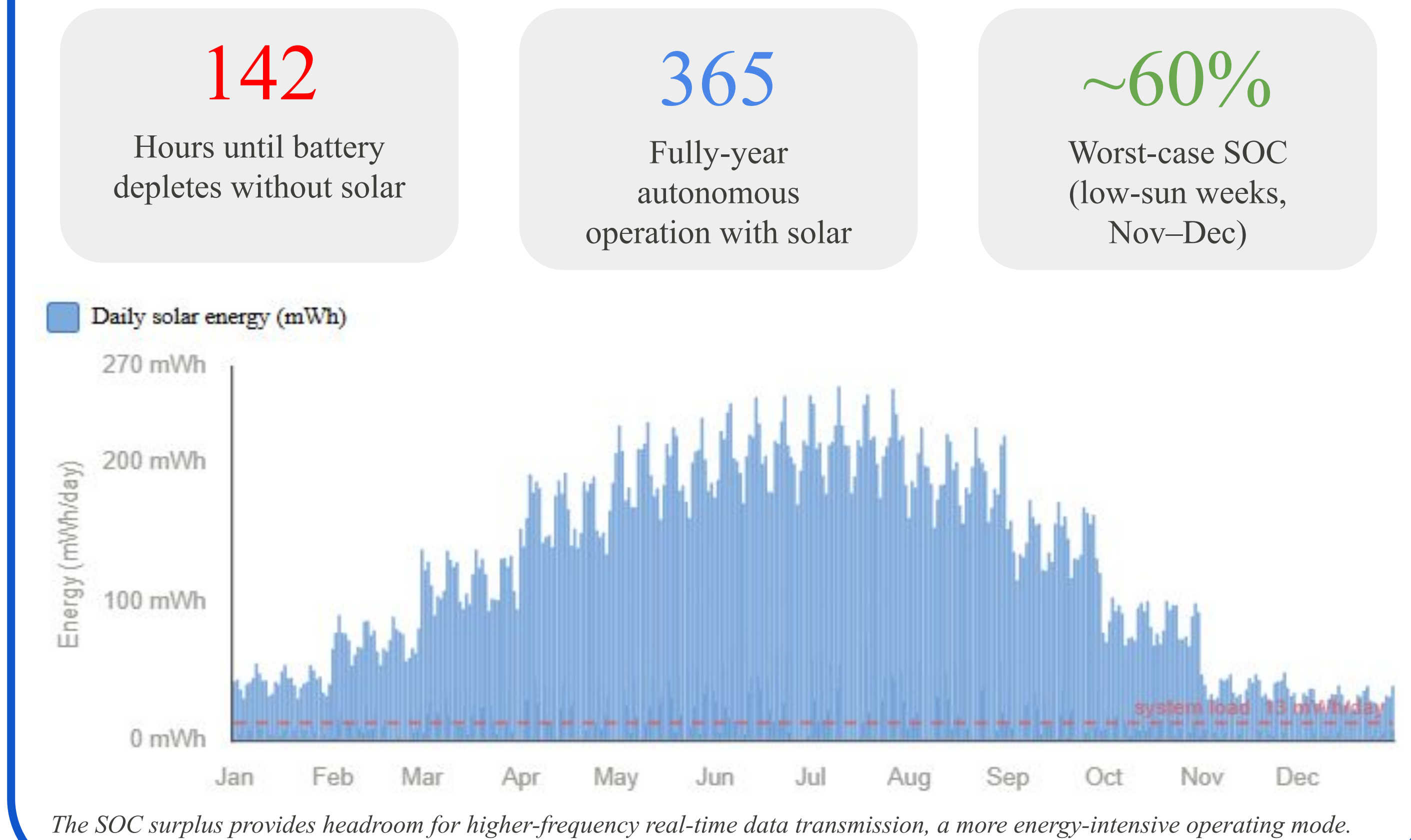
- ❖ Cup anemometer - Hall effect + magnet: RPM → wind speed via calibration factor
- ❖ Wind vane - Magnetic encoder: 0–360° direction referenced to GPS heading
- ❖ 1/4" stainless steel shaft in fixed 1/2" sleeve
- ❖ 3× Delrin bushings for corrosion resistance
- ❖ 5× 3D-printed PLA components



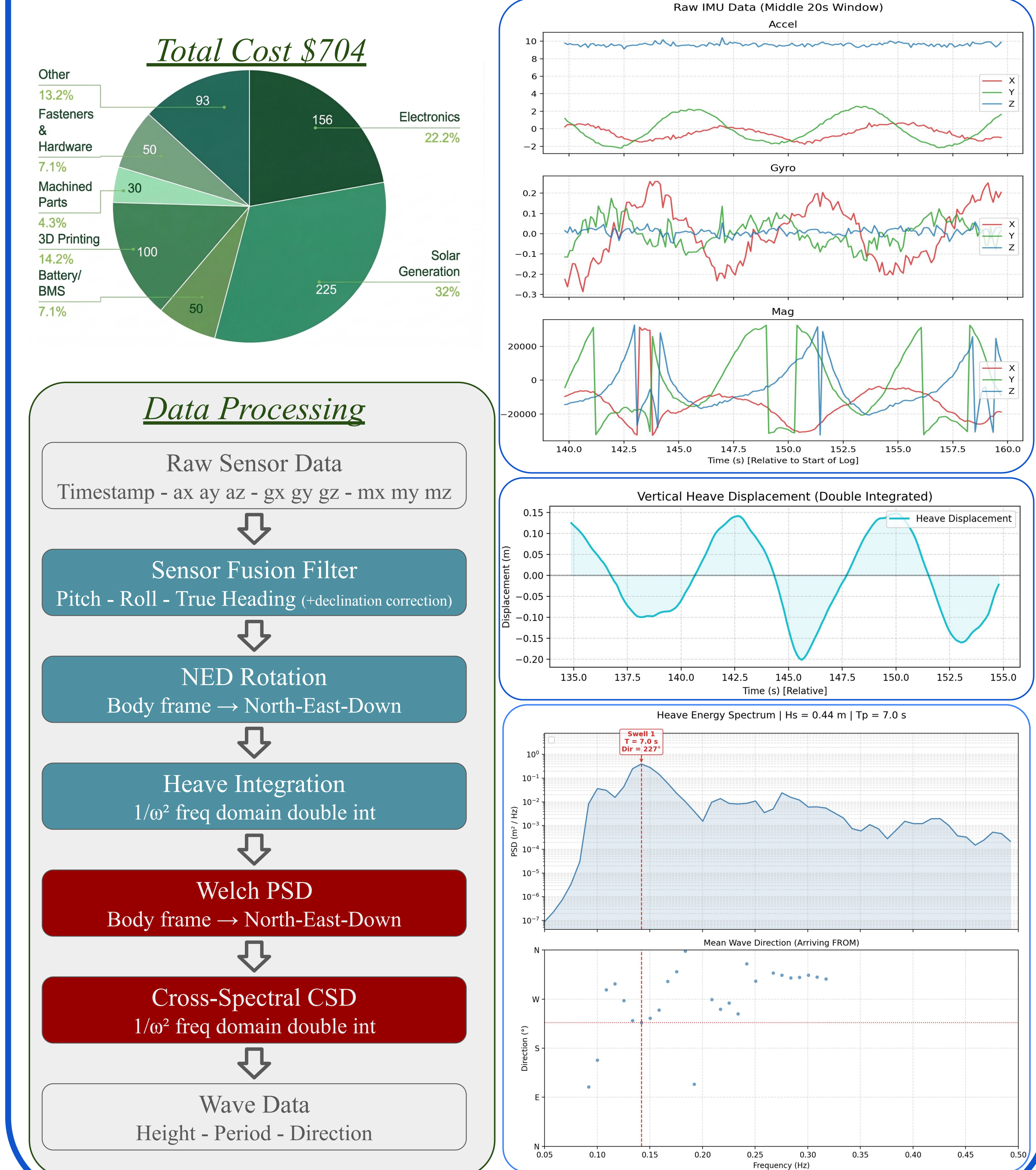
Electrical System



Energy Analysis

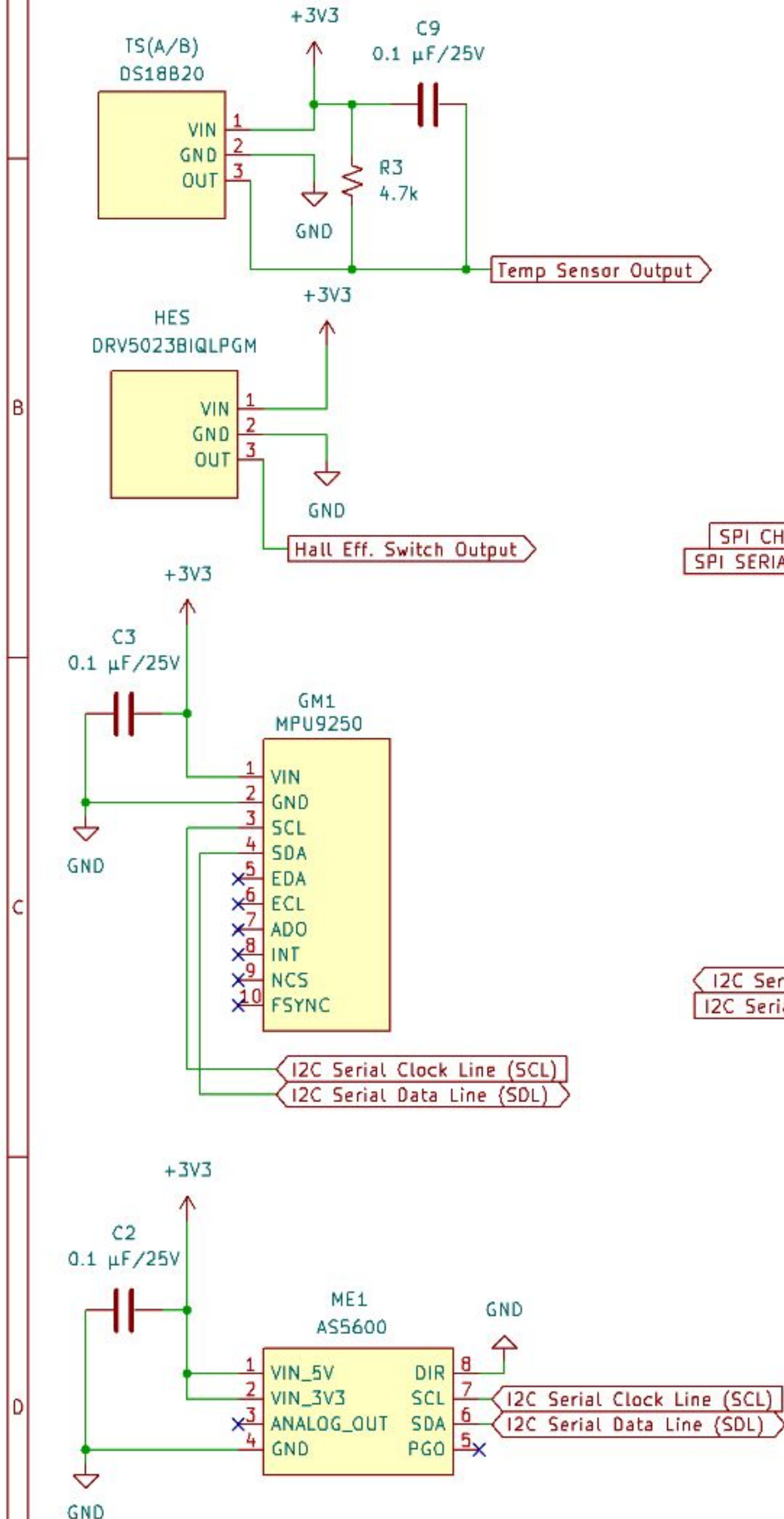


Results

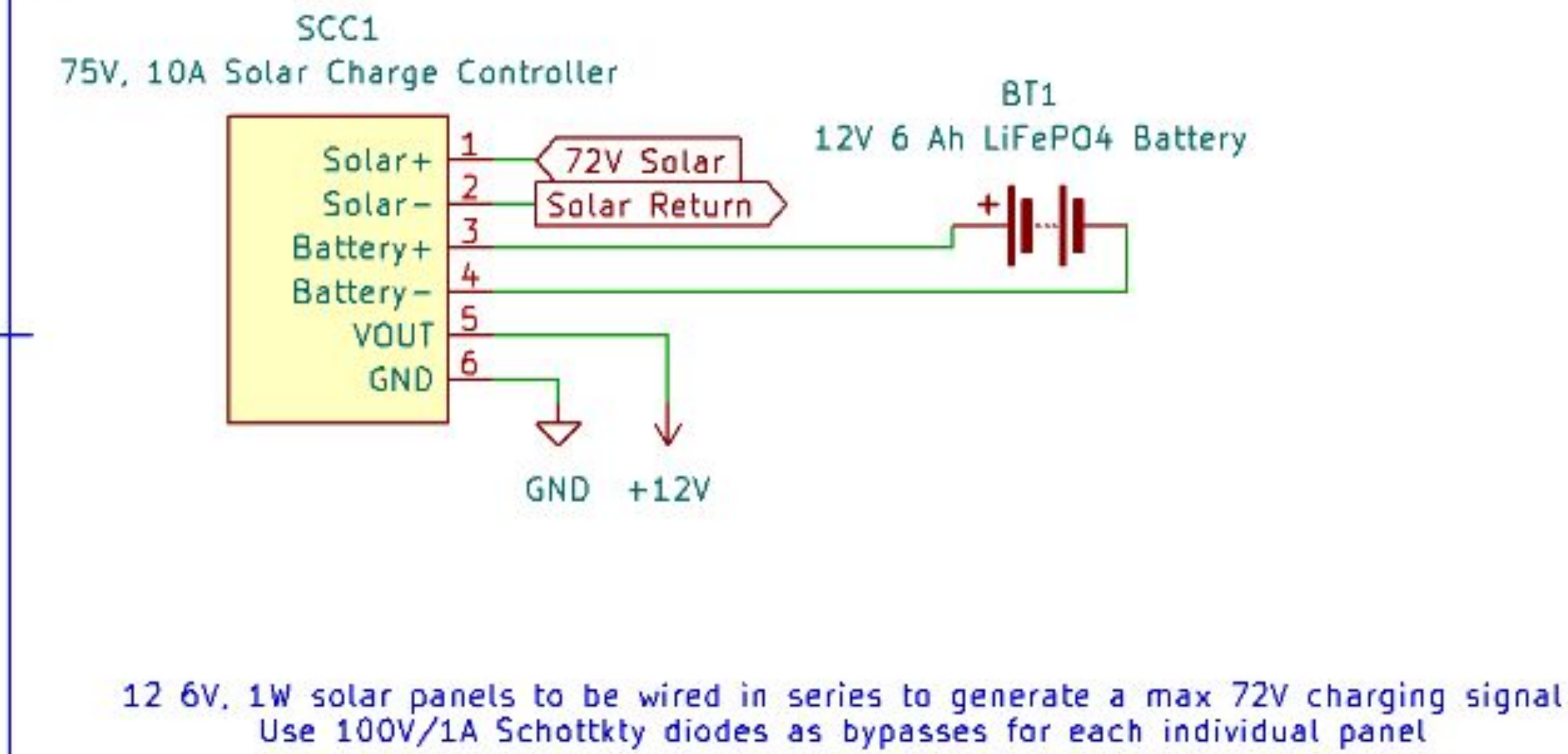


Maximum Power Usage:		
Raspberry Pi Pico:	3.3V*100mA	330mW
AS5600 Magnetic Encoder:	3.3V*6.5mA	21.5mW
MPU9250 Gyroscope Module:	3.3V*4mA	13.2mW
QMC5883P Magnetic Sensor:	3.3V*1mA	3.3mW
DRV5023BIQLPGM Hall Effect Switches:	3.3V*25mA*2	165mW
DS18B20 Temp Sensors:	3.3V*5mA*2	33mW
Solar Charge Controller:	1W*0.02	20mW
LM2596S DC-DC Buck Convrt:	1W*0.10	100mW
SD Card Reader:	3.3V*20mA	66mW
SUM =		0.75W

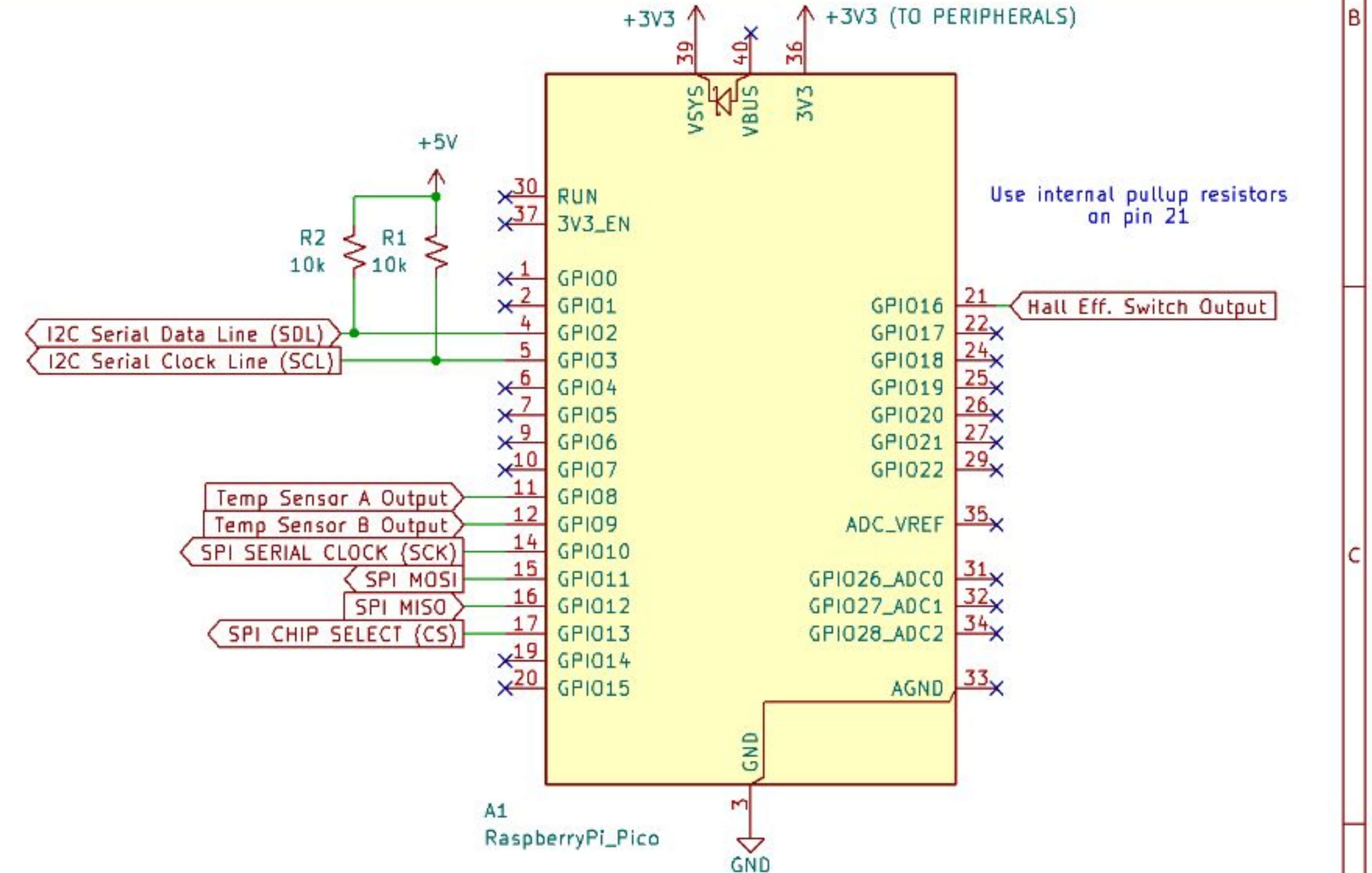
PERIPHERALS



POWER DELIVERY



MCU



Group 15

Sheet: /
File: Initial Electrical.kicad_sch

Title: MECH 400B LCSSB Electrical Schematic

Size: A4 Date: 2026-03-13
KiCad E.D.A. 9.0.7

Rev: 2
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